

Executive Summary

Our project estimates the tax adjustment amount, or the size of an IRS correction, using a simple and interpretable linear model built with the tools we learned in class. We worked from the “Suspicious Tax Returns” dataset, which includes filing details such as business expenses, tax credits, and occupation, with a flag for Small Business Owner. Before modeling, we looked at distributions and basic relationships using progress plots, and we kept a short correlation view focused on the target and a few core predictors. We noticed early that returns from SBOs seemed to behave differently, especially around credits and home-office deductions. This was an observation that also matched the dataset overview provided.

Guided by that EDA, we started with a broad specification, and then added interactions where the patterns clearly split by occupation. The final model includes business expenses, home office deduction, tax credits, income variability, a level shift for SBO, and two targeted interactions. The result is a compact model that utilizes tools learned in class and remains easy to explain. In terms of fit, the model achieves an Adjusted R^2 of 0.8528, which suggests the chosen factors capture most of the variation we see in the dataset.

We checked the usual assumptions using diagnostics. The residuals vs. fitted plot is centered without a strong curve, which supports linearity at our level. The residual histogram looks roughly symmetric with only mild tails, which is acceptable for normality. The residuals by occupation comparison shows wider spread for SBO filers. Interpreting the coefficients gives straightforward, practical takeaways. For example, holding other variables constant, every +\$1,000 in business expenses is associated with about +\$189.56 more in IRS adjustments. For home office deduction, the story splits by occupation: for non-SBO filers, +\$1,000 corresponds to about -\$463.05, but once we include the SBO interaction, the total effect for SBOs flips strongly positive to about +\$10,802.21 per \$1,000.

These results point to clear implications. Larger business expenses and certain deductions/credits are tied to larger corrections, and the relationships are noticeably stronger for Small Business Owners. From a practical standpoint, that suggests SBO filers may need closer documentation around credits and home-office claims, and reviewers may focus attention there first. At the same time, our model is linear and based on one dataset; the findings are associational. We also acknowledge the variance is higher in the SBO group. Within our course scope, we prioritized clarity and stayed with the basic approach rather than layering on advanced fixes. Finally, we view these results as a guide for better documentation and review, not a reason to profile any occupation.

In short, a small, interpretable model that adds two SBO interactions explains roughly 85% of the variation in IRS adjustments and highlights a few high-leverage levers including expenses, credits, and home-office deductions, where careful reporting may matter most—especially for Small Business Owners.